Bird Abundance

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Topic, Reserach Questions, and Ideas for Analysis

Topic:

The topic we seek to investigate is the overall variation in bird species abundance by season based on water elevation, and variation in species abundance by season between taxonomic groups.

Question(s):

- How does water elevation effect bird species abundance within the Waterfowl and Shorebird taxonomic groups across seasons? **Addressing Rubric Feedback**: focusing on one, more specific, research question as opposed to three similar questions.

Ideas for analysis and background

- Plot showing total shorebird counts by species in 2025 water year
- Plot showing total waterfowl counts by species in 2025 water year
- Plot showing trends in water elevation by date in 2025 water year
- Plot showing waterfowl and shorebird species counts across water elevations in 2025 water year, 1 plot for wy 2025, 4 faceted plots by season.

Background

Visualizations

1. Set up

Packages

```
# general use
library(tidyverse)
library(here)
library(janitor)
library(snakecase)
library(dplyr)

#Wrap Axis Label
library(scales)

#reading .xlsx files
library(readxl)

# visualize missing data
library(naniar)

# visualizing
library(patchwork)
library(flextable)
library(ggridges)
library(NatParksPalettes)
library(paletteer)
library(ggrepel)
library(cowplot)
```

Data

```
birds <- read_csv(here("data", "birds.csv"))

venoco_water <- read_csv(here("data", "venoco-water.csv"))
```

2. Cleaning

Wrangling birds

```r
#created a new object from birds
bird_wy2025 <- birds %>%
  # exclude repeat observations
  mutate(observation_date = as.Date(observation_date, format = "%m/%d/%y"))
   ↪  %>%
  filter(observation_date > as.Date("2024-11-13") & observation_date <
   ↪  as.Date("2025-10-22"),
         repeat_observation == "No",
         e_bird_group %in% c("Waterfowl",
                             "New World Sparrows",
                             "Martins and Swallows",
                             "Shorebirds")) %>%
  # filling in taxanomic information about order and families for eBird
   ↪  groups
  mutate(lowest_group = case_when(
    e_bird_group == "New World Sparrows" ~ "Passerellidae",
    e_bird_group == "Martins and Swallows" ~ "Hirundinidae",
    e_bird_group == "Shorebirds" ~ "Charadriiformes",
    e_bird_group == "Waterfowl" ~ "Anatidae"
  ))



# creating new object from most_abundant_groups
abundant_species_counts <- bird_wy2025 %>%
  # grouping by speices
  group_by(season, e_bird_group, species) %>%
  # calculating total counts for each species
  summarize(total_count = sum(count)) %>%
  # arranging in descending order
  arrange(-total_count)
```

Wrangling Water

```r
# Creating a new object from venoco_water
venoco_clean <- venoco_water |>
```

```
  # filter to only include 2025 water year
  filter(wy == "2025") |>
  # creating a new column to convert water elevation in ft to meters
  mutate(comp_level_m = (comp_level_ft + 2.96) * 0.3048) |>
  # selecting columns of interest
  select(date, comp_level_m, temperature)


# create a new object from Venoco_clean
venoco_daily <- venoco_clean |>
# group by date
  group_by(date) |>
# calculate mean water level for each day
  summarize(mean_level_m = mean(comp_level_m, na.rm = TRUE)) |>
# ungroup the data frame
  ungroup()
```

Shorebird Wrangling

```
shorebird_total_count <- bird_wy2025 |>
  #filter to only include herons, ibis and allies
  filter(lowest_group == "Charadriiformes") |>
  #group by species
  group_by(species) |>
  #calc total counts per species
  summarize(total_count = sum(count)) |>
  #arrange in descending order by total counts
  arrange(-total_count)
```

Waterfowl Wrangling

```
waterfowl_total_count <- bird_wy2025 |>
  #filter to only include herons, ibis and allies
  filter(lowest_group == "Anatidae") |>
  #group by species
  group_by(species) |>
  #calc total counts per species
  summarize(total_count = sum(count)) |>
```

```
  #arrange in descending order by total counts
  arrange(-total_count)
```

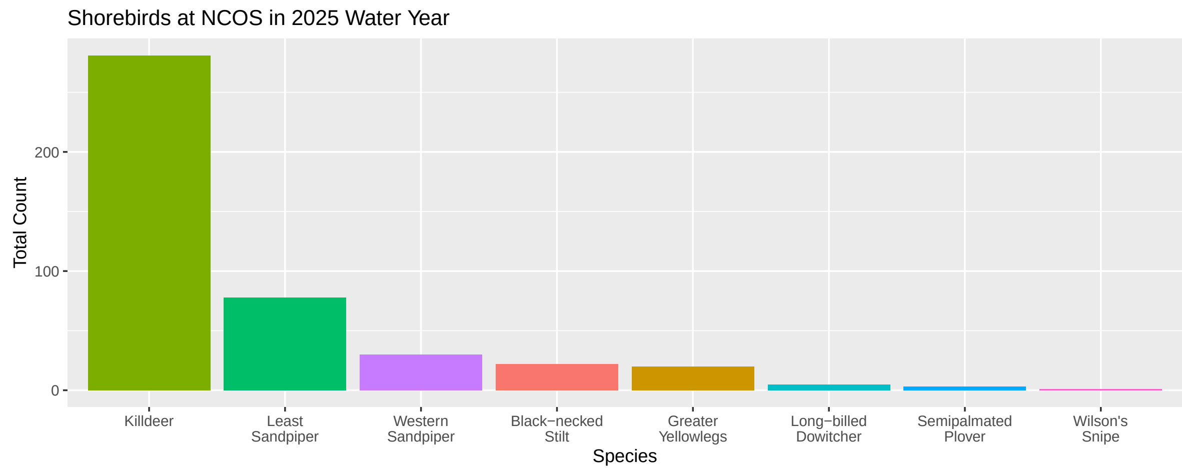
Wrangling Shorebird and Waterfowl

```
shorebird_waterfowl_25 <- bird_wy2025 |>
  #filter to only include herons, ibis and allies
  filter(e_bird_group %in% c("Waterfowl", "Shorebirds")) |>
  #group by species
  group_by(e_bird_group, observation_date) |>
  #calc total counts per species
  summarize(total_count = sum(count)) |>
  ungroup() |>
  left_join(venoco_daily,
  # naming shared columns between both data frames
  by = c("observation_date" = "date"))
```

3. Exploring Visualizations

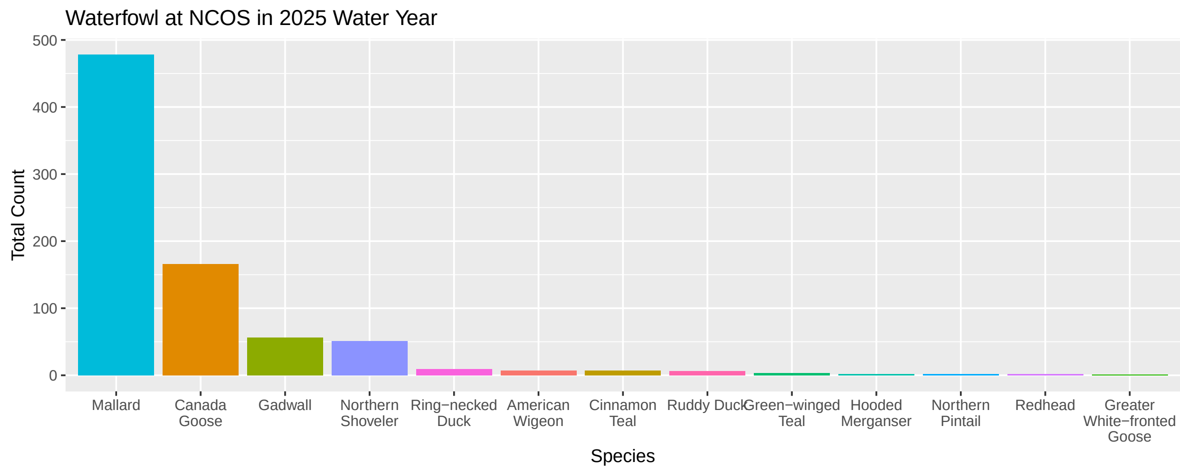
Plotting Shorebirds

```
ggplot(data = shorebird_total_count,
       # x axis is species in descending order of total count
       # # y axis is total count
       # filling columns by species
       mapping = aes(x = reorder(species, -total_count),
                     y = total_count,
                     fill = species)) +
  #first layer: column to represent counts
  geom_col() +
  # taking out the legend
  theme(legend.position = "none") +
  #relabelling x and y axes, adding title
  labs(x = "Species",
       y = "Total Count",
       title = "Shorebirds at NCOS in 2025 Water Year") +
  scale_x_discrete(labels = ~ str_wrap(., width = 10))
```



Plotting Waterfowl

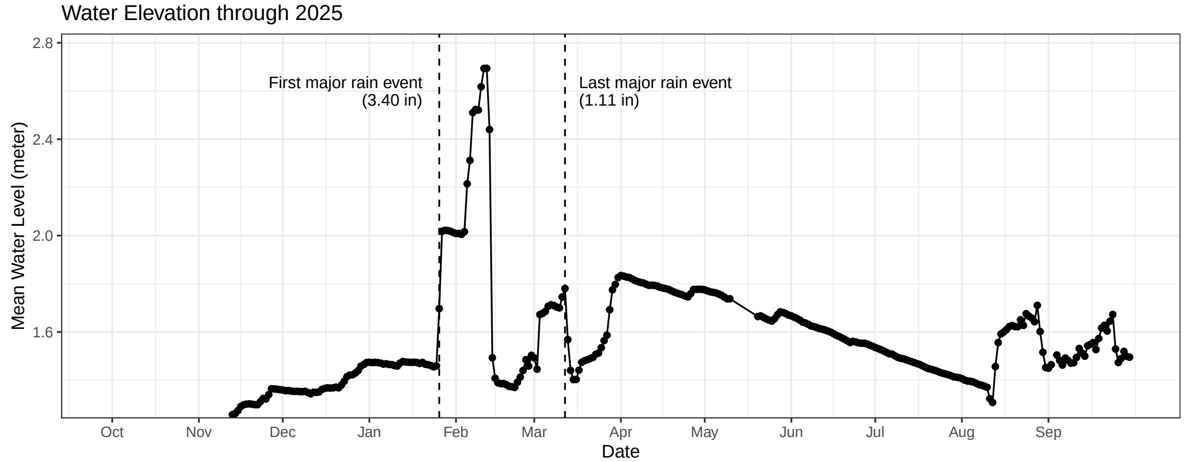
```
ggplot(data = waterfowl_total_count,
       # x axis is species in descending order of total count
       # # y axis is total count
       # filling columns by species
       mapping = aes(x = reorder(species, -total_count),
                     y = total_count,
                     fill = species)) +
  #first layer: column to represent counts
  geom_col() +
  # taking out the legend
  theme(legend.position = "none") +
  #relabelling x and y axes, adding title
  labs(x = "Species",
       y = "Total Count",
       title = "Waterfowl at NCOS in 2025 Water Year") +
  scale_x_discrete(labels = ~ str_wrap(., width = 10))
```



Plotting Water Elevation of Time 2025

```
ggplot(data = venoco_daily,
       # x axis is species in descending order of total count
       # # y axis is total count
       # filling columns by species
       mapping = aes(x = date,
                     y = mean_level_m)) +
  #first layer: column to represent counts
  geom_point() +
  # adding line
  geom_line() +
  # adjusting appearance of x-axis
  scale_x_date(
    # setting limits (no upper bound)
    limits = c(as_date("2024-10-01"), NA),
    # setting breaks so that date labels appear every month
    breaks = as_date(c("2024-10-01", "2024-11-01", "2024-12-01",
                       "2025-01-01", "2025-02-01", "2025-03-01",
                       "2025-04-01", "2025-05-01", "2025-06-01",
                       "2025-07-01", "2025-08-01", "2025-09-01")),
    # label dates with month abbreviations
    date_labels = "%b") +
   scale_y_continuous(expand = expansion(mult = c(0.01, 0.1))) +
  # custom theme
  theme_bw() +
  # taking out the legend
```

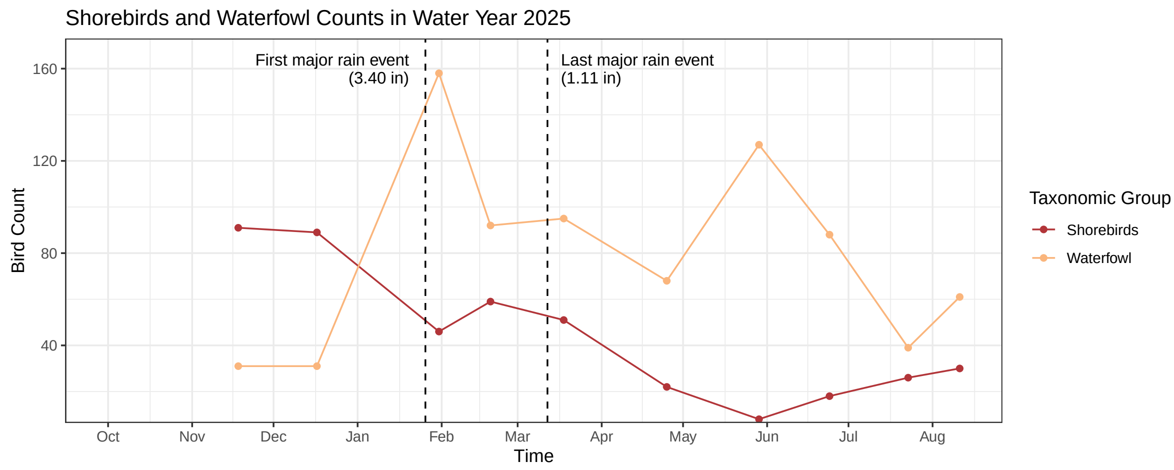
```
theme(legend.position = "none") +
#relabelling x and y axes, adding title
labs(x = "Date",
     y = "Mean Water Level (meter)",
     title = "Water Elevation through 2025") +
# add vertical line for major rain event in January
geom_vline(xintercept = as_date("2025-01-26"),
           linetype = "dashed") +
# text label (from cowplot package)
draw_label("First major rain event\n(3.40 in)",
           # x- and y-axis position
           x = as_date("2025-01-20"),
           y = 2.6,
           # right alignment
           hjust = 1,
           # text size
           size = 10) +
# add vertical line for major rain event in March
geom_vline(xintercept = as_date("2025-03-12"),
           linetype = "dashed") +
# text label
draw_label("Last major rain event\n(1.11 in)",
           # x- and y-axis position
           x = as_date("2025-03-17"),
           y = 2.6,
           # left alignment
           hjust = 0,
           # text size
           size = 10)
```



Plotting Shorebirds and Waterfowl counts through time 2025 WY

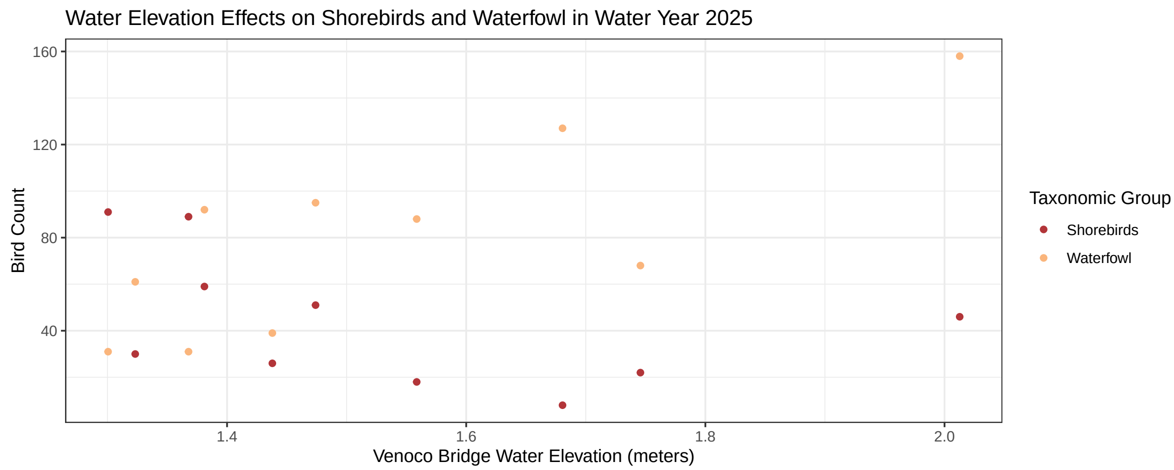
```
ggplot(data = shorebird_waterfowl_25,
       aes(x = observation_date,
           y = total_count,
           color = e_bird_group)) +
  # setting point geometry
  geom_point() +
  # adding line geometry
  geom_line() +
   # adjusting appearance of x-axis
  scale_x_date(
    # setting limits (no upper bound)
    limits = c(as_date("2024-10-01"), NA),
    # setting breaks so that date labels appear every month
    breaks = as_date(c("2024-10-01", "2024-11-01", "2024-12-01",
                       "2025-01-01", "2025-02-01", "2025-03-01",
                       "2025-04-01", "2025-05-01", "2025-06-01",
                       "2025-07-01", "2025-08-01", "2025-09-01")),
    # label dates with month abbreviations
    date_labels = "%b") +
   scale_y_continuous(expand = expansion(mult = c(0.01, 0.1))) +
  # changing background theme
  theme_bw() +
  # custom colors
  scale_color_paletteer_d("nationalparkcolors::DeathValley") +
  # Adding axis titles, title and legend
```

```r
labs(
  x = "Time",
  y = "Bird Count",
  color = "Taxonomic Group",
  title = "Shorebirds and Waterfowl Counts in Water Year 2025") +
# add vertical line for major rain event in January
geom_vline(xintercept = as_date("2025-01-26"),
           linetype = "dashed") +
# text label (from cowplot package)
draw_label("First major rain event\n(3.40 in)",
           # x- and y-axis position
           x = as_date("2025-01-20"),
           y = 160,
           # right alignment
           hjust = 1,
           # text size
           size = 10) +
# add vertical line for major rain event in March
geom_vline(xintercept = as_date("2025-03-12"),
           linetype = "dashed") +
# text label
draw_label("Last major rain event\n(1.11 in)",
           # x- and y-axis position
           x = as_date("2025-03-17"),
           y = 160,
           # left alignment
           hjust = 0,
           # text size
           size = 10)
```



Plotting Shorebirds and Waterfowl counts over different Water Elevations

```
ggplot(data = shorebird_waterfowl_25,
       aes(x = mean_level_m,
           y = total_count,
           color = e_bird_group)) +
  # setting point geometry
  geom_point() +
  # changing background theme
  theme_bw() +
  # custom colors
  scale_color_paletteer_d("nationalparkcolors::DeathValley") +
  # Adding axis titles, title and legend
  labs(
    x = "Venoco Bridge Water Elevation (meters)",
    y = "Bird Count",
    color = "Taxonomic Group",
    title = "Water Elevation Effects on Shorebirds and Waterfowl in Water
    ↪  Year 2025")
```



Addressing Rubric Feedback: water elevation is plotted with abundance using a scatter plot rather than a bar chart. Additionally, we are now comparing our response variable across bird groups instead of isolating their patterns.

Elective Planning

Idea: Illustrated informational bird species pamphlet for bird watchers, with insights on seasonal species abundance trends and ecological significance.

Drafts description/image

ElectivePlanPDF

Plan for making progress in the next two weeks

- Gather resources for information on most abundant species within the shorebird and Waterfowl taxonomic groups
  - KilldeerPDF
  - LeastSandpiperPDF
- Draft small 'blurb' descriptions for each species to accompany illustrations
- Complete illustrations
- Compile illustrations and written text into formatted pamphlet to be printed